AI HACKATHON ROUND 1

EcoPulse AI

The Heartbeat of Smart Energy Agent

Sustaining Tomorrow with Reinforcement Learning

THE ENERGY DILEMMA

⚠ Grid Rigidity
Traditional grids cannot adapt to the sharp spikes of renewable energy in modern cities.

🌪 Peak Wastage
Renewable energy is often wasted when generation exceeds immediate local demand.

THE AI AGENT SOLUTION

Autonomous management of Storage and Consumption.

- **Adaptive Charge:** Learning when to store power.
- **Cost Reduction:** Real-time grid price arbitrage.

CORE INTELLIGENCE

Solar Forecast
Weather-based predictive planning.

Load Timing
Buffering for peak demand shifts.

Price Logic
Monitoring utility rate fluctuations.

SCALING THE IMPACT

Commercial Pilot
Optimizing GIFT City high-rise towers.

Smart Rajkot
Deployment in residential smart clusters.

SIMULATION & ACCESS

Performance Results

32%
Savings

95%
Renewable

Thank You

Building the Energy Future of Gujarat.

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